claude-windows / 96963dab-d38f-488c-af44-530ddbeefed1

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru-badminton-highlight-indexer\96963dab-d38f-488c-af44-530ddbeefed1\subagents\workflows\wf_dd531439-5e0\agent-a89b73ee17a7ad1cf.jsonl`
- SHA-256: `9501fb9a39378e2ca000e8a4e292ff2bd7f8b032696a534b43142cd67aa1b6d6`
- Source modified: `2026-07-08T06:37:17+00:00`
- Imported at: `2026-07-08T15:59:27+00:00`
- project: `wf_dd531439-5e0`
- session_id: `96963dab-d38f-488c-af44-530ddbeefed1`

Transcript

1. user (2026-07-08T06:36:07.112Z)

```
You are reviewing an UNCOMMITTED diff in an isolated git worktree at
C:/Users/avidu/AppData/Local/Temp/claude/wt524-969 (branch feat/524-l4-primary, based on
origin/master).
To see the change: cd to the worktree and run `git --no-pager diff` (working tree vs HEAD) and
`git --no-pager diff --stat`; read the new files
docs/COMPUTE_DECOUPLED_SERVING/L4_PRIMARY_TOPOLOGY.md and tests/test_video_id_trust.py directly.
Read surrounding code as needed ? do NOT limit yourself to diff hunks.

CONTEXT ? GitHub issue #524 (epic): collapse the pipeline onto the L4 worker; shrink control-
plane to a thin router. This PR delivers (1) a design doc resolving the compute-topology
question (recommend co-located scale-to-zero Cloud Run Jobs + bucket upload; REJECT upload-
onto-L4 and persistent-GPU with a cost model), and (2) "win #2": the control-plane drain trusts
the SERVER-DERIVED submit-time md5 (jobs.video_id ? submit_time_video_id, #484 ? GCS metadata
md5 converted to hex by backend/storage/gcs.py::_md5_hex) instead of re-hashing the fetched
clip, gated by deployment.trust_submit_video_id (default True) and guarded by
orchestration._looks_like_video_id (strict 32-char lowercase hex). Win #1 (validation-on-L4) is
deliberately NOT in this PR ? it is delivered by the separate open branch feat/521-compile-on-l4
(phase_placement.validation); duplicating it is forbidden.

Report ONLY concrete, actionable findings with file:line. No praise, no restating the design.
For each: what breaks, the exact failing input/state, and the fix. If a dimension is clean, say
so explicitly.
A prior reviewer raised this finding on the REPO CONVENTIONS / DISCIPLINE dimension:
  SEVERITY: nit
  FILE: docs/DOC_STATUS.md:34
  SUMMARY: New DRAFT design doc L4_PRIMARY_TOPOLOGY.md is not registered in the DOC_STATUS §3
lifecycle register.
  CLAIMED FAILURE: The register (§3, line 34) tracks lifecycle per tracked doc/project;
L4_PRIMARY_TOPOLOGY.md is a new self-tracked DRAFT design doc with its own status header but
gets no row or sub-note under the existing COMPUTE_DECOUPLED_SERVING/ row. Skipping the §2
archival due-diligence is correctly justified (this is not a terminal-state DONE/REJECTED flip),
but a newly created tracked design doc idiomatically earns a register entry so its DRAFT->DONE
lifecycle is visible at clean-slate.
  PROPOSED FIX: Add a one-line sub-note (or row) under the COMPUTE_DECOUPLED_SERVING/ register
entry noting L4_PRIMARY_TOPOLOGY.md is DRAFT/owner-review, archive-on-DONE. Optional; the doc's
own status header + the updated dir README are the primary trackers.

Adversarially VERIFY it against the real code/doc in
C:/Users/avidu/AppData/Local/Temp/claude/wt524-969. Default to REFUTED unless you can construct
the concrete failing input/state and confirm it in the actual code. A finding is CONFIRMED only
if it is a real defect a maintainer must fix (not a style preference, not already-handled, not a
misread). State the verdict and the evidence (file:line you checked).
```

2. assistant (2026-07-08T06:36:10.640Z)

I'll verify this finding by examining the actual DOC_STATUS.md file and the new design doc in the worktree.

3. user (2026-07-08T06:36:21.301Z)

```
warning: in the working copy of 'docs/COMPUTE_DECOUPLED_SERVING/L4_PRIMARY_TOPOLOGY.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tests/test_video_id_trust.py', LF will be replaced by CRLF the next time Git touches it
 backend/api/job_worker.py                          |   5 +
 backend/config/models.py                           |  10 +
 backend/orchestration.py                           |  51 ++++-
 .../L4_PRIMARY_TOPOLOGY.md                         | 225 ++++++++++++++++++++
 docs/COMPUTE_DECOUPLED_SERVING/README.md           |   2 +
 tests/test_config_deployment.py                    |   2 +
 tests/test_video_id_trust.py                       | 173 ++++++++++++++++
 7 files changed, 466 insertions(+), 2 deletions(-)
```

4. user (2026-07-08T06:36:22.113Z)

```
1	# Documentation Status & Archival Register
2	
3	> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-07-02.
4	> **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.
5	> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).
6	
7	---
8	
9	## 0. How to use this
10	- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
11	- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
12	- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.
13	
14	## 1. Lifecycle legend
15	`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).
16	
17	## 2. Archival due-diligence checklist  ? *the repeatable process (do NOT just `git mv`)*
18	Before a foundational/tracked doc moves to `docs/archives/`:
19	1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
20	2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21	3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22	4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
23	5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
24	6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25	
26	> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
```

WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28 ## 3. The register
29
30 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
31 | Doc | Status | Archive-ready? | Next action |
32 |---|---|---|---|
33 | `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates. A0?A9 DONE; A1 live (Cloud Run, e2e runlog); **A13 review DELIVERED 2026-07-04** (`ALPHA_LAUNCH_REVIEW_2026-07-04.md`) | No | Owner: record GO/NO-GO (§10) + close/defer A10/A11/A12; fix khelsutra #154 copy blockers; run the 5-min browser-CUJ checklist (review §1.1) |
34 | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |
35 | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the R-series; complete when §7.1 single quality number is attributed |
36 | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
37 | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |
38 | `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |
39 | `RALLY_DETECTION_FIXES_PLAN.md` | **DONE** ? findings A/B/C resolved; P2b promoted `inactivity_temporal_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + §7.1 number) | Archive with the R-series when the track wraps |
40 | `RALLY_DETECTION_CODE_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |
41 | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |
42 | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |
43 | `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |
44 | `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |
45 | `EXECUTION_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |
46 | `GOLDEN_REGRESSION_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |
47 | **Data layer** ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`. |
48 | `archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md` **(umbrella)** | **? COMPLETE / ARCHIVED (2026-07-02)** ? the five-repo code audit + folded-in *Remediation tracker* (P0?P28 all closed) + absorbed prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues (#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved to `archives/past_projects/` per §2. Residuals now live as GitHub issues, not in this doc. |
49 | `DECISION_SNAPSHOT_REGRESSION.md` | **DRAFT ? decisions LOCKED** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split |
50 | `DESKTOP_APP_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |
51 | `OWNED_MODEL_IMPLEMENTATION_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |
52 | `PERSONALIZATION_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB

v5 store landed) |
`UI_ALPHA_EXECUTION_PLAN.md`	ACTIVE	No	khelsutra UI alpha track
`I18N_PLAN.md`	Design (P0); **in implementation** (adapted)	No	Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated
`POST_166_...RISK_ASSESSMENT.md`	Plan for owner review	No	Milestones/risk doc

3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")

Doc	Status	Note
`PLATFORM_ARCHITECTURE.md`	Strategy/research	The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING
`COMMERCIALIZATION.md`	? LIVING	Strategy; carries WASB-C3
`VIDEO_LOCALITY_MODEL.md`	THINKING DOC	L0 fully-local = the truly-local profile
`HOSTING_PLAN.md`	PROPOSAL	**? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)**
`UI_PROPOSAL.md`	PROPOSAL (partly shipped)	UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
`STORAGE_SHARING_MODEL.md`	DRAFT (co-design pending)	?
`ANALYZERS_AND_RECONCILERS.md`	DESIGN (owner-gated)	?
`ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md`	LIVING references	Keep current; not archive targets

3c. Reports & terminal records (archive candidates once their track wraps)

Doc	Status	Note
`ALPHA_LAUNCH_REVIEW_2026-07-04.md`	**REPORT ? DELIVERED 2026-07-04** (A13) ? owner-facing alpha launch review: GO/NO-GO recommendation (conditional GO for the current 4-email allowlist, gated on khelsutra #154 copy fixes + the owner browser-CUJ checklist; NO-GO for wider invites until A10/A11 + SA least-privilege + retention), DoD scorecard, operator runbook, observability/triage incl. the 2026-07-04 live failure-mode battery, privacy/consent audit, known limits + rollback	Awaiting the owner GO/NO-GO record on `ALPHA_LAUNCH_READINESS.md` §10; archive with the alpha cluster once recorded
`REAL_FOOTAGE_VALIDATION_REPORT.md`	validation complete	Motivates R2; archive with the R-cluster
`RALLY_DETECTION_QUALITY_REPORT.md`	report	Quality snapshot; part of R-cluster
`QUALITY_ITERATIONS.md`	LIVING empirical log	**Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]])
`CODE_AUDIT_AND_TEST_HARDENING` · `HARDENING_LOOP_HANDOFF` · `DEFERRED_HARDENING_PLAN`	COMPLETED ? **finish-archived 2026-07-02** (top-level stubs removed; absorbed into the umbrella's *Prior audits*)	? Full copies under `docs/archives/*-COMPLETED-2026-06-14.md`; inbound links repointed (finding #4 resolved)

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

3e. Already archived (`docs/archives/`)

`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/` · `past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) · `past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

84 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`
85 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).
86
87 ## 4. Open due-diligence findings / actions (2026-06-21 review)
88
89 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).
90 1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
91 2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
92 3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
93 4. **? RESOLVED (2026-07-02) ? finish-archived the hardening stubs.** The three top-level pointer-stubs (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) were removed; full archived copies remain under `docs/archives/`, inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is summarized in the umbrella's *Prior audits (absorbed)*.
94 5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
95 6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).
96
97 ## 5. Maintenance
98 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
99

5. assistant (2026-07-08T06:36:24.541Z)

Now let me read the new design doc's status header and the updated dir README to see what the PR actually did.

6. user (2026-07-08T06:36:26.874Z)

1 # L4-primary pipeline topology ? collapse the pipeline onto the worker, shrink the control-plane to a thin router
2
3 > **Status:** `DRAFT` ? **owner-review** (recommends *against* the epic's proposed direction #1; see §0/§8) · **Owner:** Avi · **Created:** 2026-07-08 · **Last updated:** 2026-07-08
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
5 > **Issue:** [#524](https://github.com/Khelsutra/badminton-highlight-indexer/issues/524) (epic) · **Builds on:** [#521](https://github.com/Khelsutra/badminton-highlight-indexer/issues/521) (`phase_placement`) · **Motivated by:** [#520](https://github.com/Khelsutra/badminton-highlight-indexer/issues/520) (the tracing that surfaced the data-movement finding)
6 > **Relation to existing docs:** extends

[COMPUTE_DECOUPLED_SERVING/README.md](./README.md) (ADR-014) and [`PHASE_PLACEMENT.md` (#521)](https://github.com/Khelsutra/badminton-highlight-indexer/blob/feat/521-compile-on-l4/docs/COMPUTE_DECOUPLED_SERVING/PHASE_PLACEMENT.md); it does **not** supersede them ? it answers the *topology* question they left open and sequences the migration on top of `phase_placement`.

> **Honesty note:** each claim is tagged `[verified]` (checked against code/runlog), `[researched]` (needs sign-off / external pricing I could not fetch live), or `[design]` (proposed). Not legal/financial advice.

---

0. TL;DR

The 2026-07-08 live CUJ moved one 1.83 GiB clip **3×** (upload?bucket, bucket?control-plane to validate, bucket?worker to detect) and pinned the CPU-only 2-vCPU control-plane for **~16 min** of heavy work (validation 183 s + stitch ~12m55s). The epic proposes fixing this by **uploading directly onto the L4** and async-copying to the bucket. **This design recommends *against* that direction** and quantifies why: a same-region bucket?compute transfer inside GCP is **LAN-fast and not billed as internet egress** `[researched]`, so the round-trip that upload-onto-L4 saves is worth **~seconds and ~$0**, while keeping an L4 up to *receive* a **minutes-long** upload burns **~175 s of idle GPU per run (~$0.08?0.16)** `[design]` and *opens* a durability exposure window instead of closing one.

The cheaper, more durable, already-ratified path is the opposite: **keep the upload streaming to the cheap bucket** (the #480 design ? the durable copy exists *before* the L4 ever starts), and **collapse validate + detect + stitch onto the scale-to-zero Cloud Run L4 Job** ? which is exactly what **#521's `phase_placement` already wires as a config flip**. The residual "~30 s pre-validation gap" is then killed not by moving the *upload* but by the control-plane **never downloading the file at all** (it delegates validation to the L4 per #521, and trusts the already-known submit-time md5 per this epic's win #2 ? no re-hash). Net data movement drops from 3× to the theoretical floor: **1 client upload + 1 LAN-fast intra-region pull**, with **zero idle-GPU cost** and **zero exposure window**.

Actionable now (this PR): the design doc + **win #2** (drop the redundant control-plane re-hash, config-guarded, reversible). **Win #1** (validation-on-L4) is **delivered by #521** ? this doc does not duplicate its knob. A persistent GPU (GCE VM / Cloud Run Service-with-GPU) is **rejected at current volume** on a quantified cost basis and left as an owner-gated future ADR.

1. Problem & goal

Why now. Tracing the live CUJ (`runlogs/DEPLOY_REPORT_cloud-serving_validation-latency_2026-07-07.md` §9 `[verified]`) produced two findings:

1. **Data is moved 3× per run** `[verified]`: (a) upload streams straight to `gs://?/uploads/?` (never touches control-plane disk ? the #480/#471 OOM fix, `backend/api/uploads.py`), computing the md5 incrementally as it streams; (b) the control-plane re-downloads the whole clip to scratch to validate ? `_execute_processing` ? `storage.acquire_local_input(gs://?)` at `backend/orchestration.py:748`, then a **redundant re-hash** `compute_video_id` at `:751`; (c) the worker downloads it a third time to detect (`[worker] pulled ?`). The unlogged **~30 s pre-validation gap** is finding (b).
2. **The control-plane is the scaling wall** `[verified]`: validation (183 s) **and** stitch (~12m55s, ~67 s/clip 4K re-encode, no NVENC) both run in-process on the 2-vCPU control-plane, serialized by `_LOCAL_COMPUTE_LANE` ? ~16 min of heavy CPU per run ? the API saturates past ~5 concurrent users.

The concrete outcome we want. The L4 worker becomes the primary compute for the whole pipeline; the control-plane shrinks to a **thin router** (auth · dispatch · DB · SPA + job status · signed-URL download) that does **no heavy decode/encode** and, ideally, **never pulls the full file**. Every move must be **reversible by config** with a **bucket-mediated fallback**, and land as small PRs on top of `origin/master`.

The hard question this doc must resolve (compute topology). "Upload directly to the L4" needs an **inbound endpoint on the GPU box**, but the current L4 is a Cloud Run **Job** (batch, no inbound HTTP ? `backend/compute/cloud_run.py`, dispatched per-request, `[verified]`).

So the epic forces a choice between **A. Cloud Run Service-with-GPU**, **B. GCE L4 VM**, and **C. a hybrid** presign/redirect. §4 answers it with numbers.

Read to get current: [`PHASE_PLACEMENT.md` (#521)](https://github.com/Khelsutra/badminton-highlight-indexer/blob/feat/521-compile-on-l4/docs/COMPUTE_DECOUPLED_SERVING/PHASE_PLACEMENT.md) ? the `phase_placement` knob this sequences on, [README.md](./README.md) §2 decisions **D4/D7/D16** (the ratified constraints that bound the options), and the runlog §9 (the evidence).

2. Decisions locked

#	Decision	Source / date	Implication
L1	**Keep upload ? bucket** (streaming, #480). The durable copy exists **before** any compute runs; the bucket *is* the "reliable co-located disk" the epic wants ? no VM needed for durability.	this doc `[design]` + #480 `[verified]`	No exposure window; upload cost stays on the cheap I/O path.
L2	**Collapse validate + detect + stitch onto the scale-to-zero Cloud Run L4 Job** via `phase_placement` (#521), not onto a persistent GPU. Honors **D7** (scale-to-zero) + **D16(a)** (Jobs, not Services).	#521 `[verified]` + D7/D16	The "L4-primary" vision is realized by **config flips**, not a rearchitecture.
L3	**Kill the ~30 s gap by *not downloading to the control-plane*, not by uploading onto the L4.** Compose #521's validation-delegation (CP skips the validator suite) with win #2 (CP trusts the submit-time md5 ? no re-hash).	this doc `[design]`	The CP stops pulling the full file when validation is delegated; movement ? 3×?2×.
L4	**Reject persistent-GPU (GCE VM / Service-with-GPU) at current volume.** Breakeven vs scale-to-zero is a **~51 % on-demand / ~31 % spot duty cycle** (? 2,600?4,300 runs/mo sustained) `[design]`; we are at pilot volume. Revisit only behind a **new ADR** at sustained load.	§4.2 cost model `[design]`	Preserves per-run billing (D7/D8); avoids ~$382?637/mo idle.
L5	**Reject "upload onto the L4."** It saves a ~free, seconds-fast intra-region LAN pull at the cost of ~175 s idle GPU/run **and** opens a durability window.	§4.3 `[design]`	The epic's proposed direction #1 is not adopted; owner sign-off requested (§8).

7. user (2026-07-08T06:36:34.068Z)

```
diff --git a/docs/COMPUTE_DECOUPLED_SERVING/README.md b/docs/COMPUTE_DECOUPLED_SERVING/README.md
index e60d644..d0f3fea 100644
--- a/docs/COMPUTE_DECOUPLED_SERVING/README.md
+++ b/docs/COMPUTE_DECOUPLED_SERVING/README.md
@@ -11,6 +11,8 @@

 > **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13, a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user *confirms* (D15).

+> **#524 (2026-07-08):** the **L4-primary topology** question ? "collapse upload?stitch onto the worker, shrink the control-plane to a thin router" ? is resolved in [L4_PRIMARY_TOPOLOGY.md](./L4_PRIMARY_TOPOLOGY.md). It **recommends against** uploading directly onto the GPU box (a cost model shows that trades a ~free intra-region LAN pull for minutes of idle GPU) and instead **collapses validate+detect+stitch onto the scale-to-zero L4 Job** via #521's `phase_placement` ? honoring D4/D7/D16. Ships **win #2** here (drop the redundant control-plane re-hash; trust the submit-time md5, reversible).
+
 ## 0. TL;DR
 The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB / mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local** (local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive / bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus **cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the
```

same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the **parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.

8. assistant (2026-07-08T06:37:10.594Z)

I have enough to render a verdict. Let me confirm the finding's core claims against what I've read.

Verified facts:
- DOC_STATUS.md §3 (register, lines 28?85) contains no row or sub-note for `L4_PRIMARY_TOPOLOGY.md`. The PR does not touch DOC_STATUS.md at all (diff stat lists 7 files; DOC_STATUS.md is not among them). So the literal claim "not registered in §3" is factually true.
- The new doc has its own status header + lifecycle (`L4_PRIMARY_TOPOLOGY.md:3-4`): `DRAFT ? owner-review`, `DRAFT ? IN PROGRESS ? DONE ? archived`.
- The PR updated the dir README (`docs/COMPUTE_DECOUPLED_SERVING/README.md`) with a #524 line pointing to the new doc.

Now whether this is a defect a maintainer must fix:

1. The register's established convention is to track `COMPUTE_DECOUPLED_SERVING/` as ONE directory-level row (line 34), NOT to enumerate its sub-docs. The sibling sub-docs in that same directory ? `README.md` and `PHASE_PLACEMENT.md` (#521) ? have no individual register rows either. The "Data layer ? docs/data_pipeline/" entry (line 47) follows the identical pattern: a whole directory tracked as one lifecycle unit via its dir README. So NOT adding a row for `L4_PRIMARY_TOPOLOGY.md` is consistent with the register's own convention, not a violation of it. The new doc is already covered by row 34 plus the updated dir README (its listed tracker).

2. §2 archival due-diligence is explicitly NOT triggered (not a terminal-state DONE/REJECTED flip) ? the finding concedes this.

3. The finding is self-labeled `nit` and the proposed fix says verbatim "Optional; the doc's own status header + the updated dir README are the primary trackers." There is no failing input/state ? the register remains internally consistent and complete under its existing directory-level tracking convention.

This is a subjective style preference the reviewer themselves marked optional, and it contradicts the register's established convention. It is not a defect a maintainer must fix.

9. user (2026-07-08T06:37:17.580Z)

Structured output provided successfully